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Biochemistry Module, 1st Year Medicine
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TRIGLYCERIDE METABOLISM

I/INTRODUCTION

- Triglycerides (TG) are the main form of lipid storage in the body.
- They consist of glycerol esterified with three fatty acids (FA) and represent a major source of energy due to their high storage capacity and high energy yield (more than six times that of glycogen on a weight-for-weight basis).

II. Roles and Forms of Triglycerides

1. Dietary intake:

- Main source of dietary fats (>90%).
- Carriers of fat-soluble vitamins (A, D, E, K) and essential fatty acids (e.g. omega-3).

2. Plasma transport:

TG are transported in the form of lipoproteins (e.g. chylomicrons, VLDL) due to their hydrophobic nature.

3. Intracellular storage:

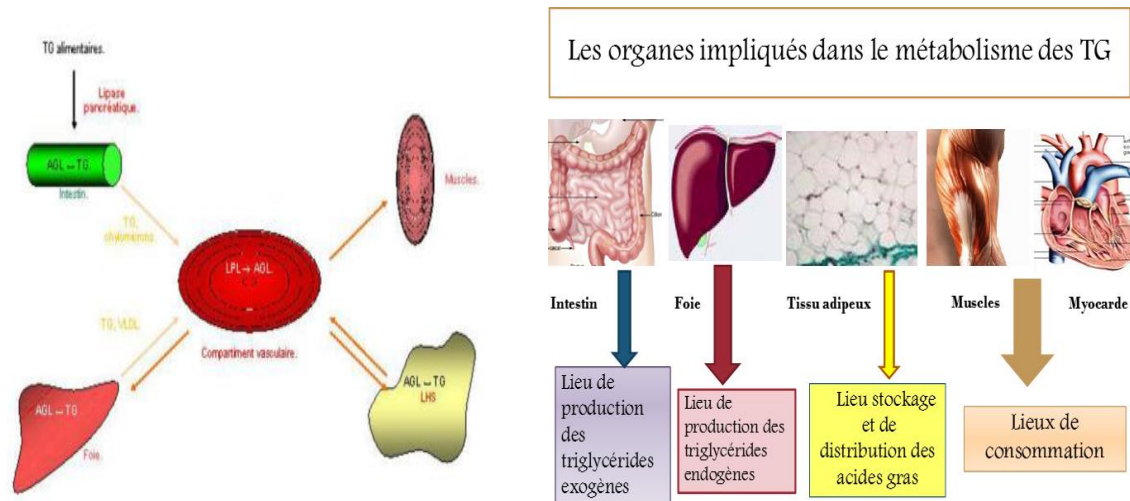
Located mainly in adipose tissue, which serves as an anhydrous energy reserve (non-hydrated, and therefore more concentrated).

✚ Why are fatty acids in the form of TG the preferred form of energy reserve?

- Anhydrous reserve: As TG are non-polar, they provide 6 times more energy than glycogen on a weight-for-weight basis.
- Reduced fatty acid carbon chain: their oxidative catabolism provides more energy than the oxidation of carbohydrates and proteins on a weight-for-weight basis.

III. Organs involved

- 4 organs involved:
 - **Digestive tract:** digestion and absorption of dietary lipids
 - **Liver:** synthesis of endogenous triglycerides from non-lipid molecules
 - **Adipose tissue:** site of energy storage and reserves in the form of fatty acids and triglycerides
 - **Peripheral tissues:** kidneys, heart and muscles



➤ Enzymes involved in TG metabolism

- **TG synthesis** is carried out by a multi-enzyme complex: **TG synthase**.

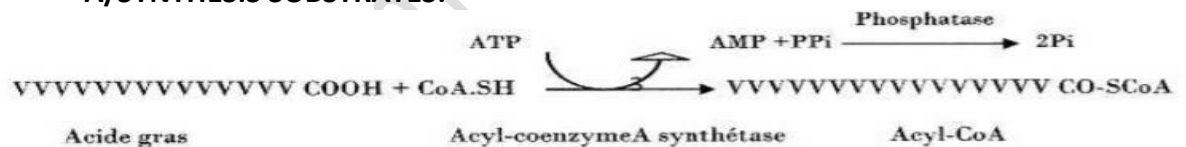
There are **two** types, depending on whether the cells are in the enterocyte, adipose tissue, muscles, myocardium or liver.

- **TG catabolism** is carried out by **three** enzymes that differ in their locations and specific functions:

- Pancreatic lipase hydrolyses dietary TG.
- Extracellular lipoprotein lipase hydrolyses circulating TG.
- Cellular TG lipase.

IV. TG BIOSYNTHESIS:

A/SYNTHESIS SUBSTRATES:



-**Fatty Acids**: activated in the form of **acyl-CoA**, a reaction catalyzed by **acyl-CoA synthase**.

-**Glycerol**: there are **two active** forms:

- **2-Monoacylglycerol**: derived from the hydrolysis of dietary triglycerides by pancreatic lipase.

- **Glycerol 3-phosphate**: two sources:

1-reduction of DHAP (from glycolysis):

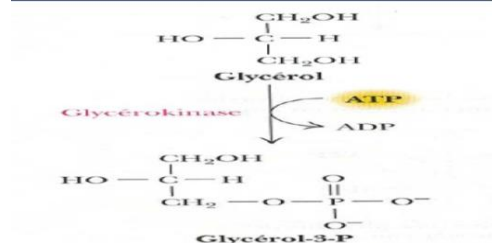
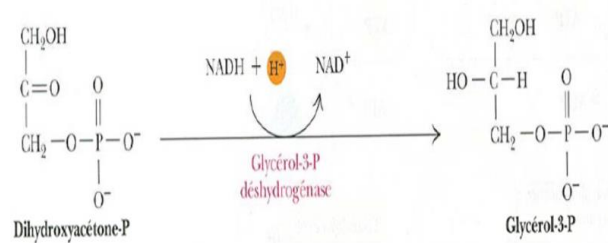
Glycerol-3-phosphate dehydrogenase



2- via phosphorylation of glycerol (from free glycerol, mainly in the liver).

Glycerol kinase



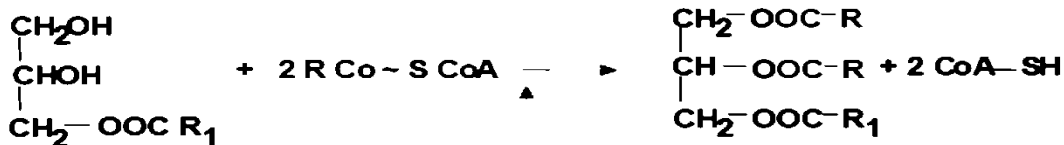
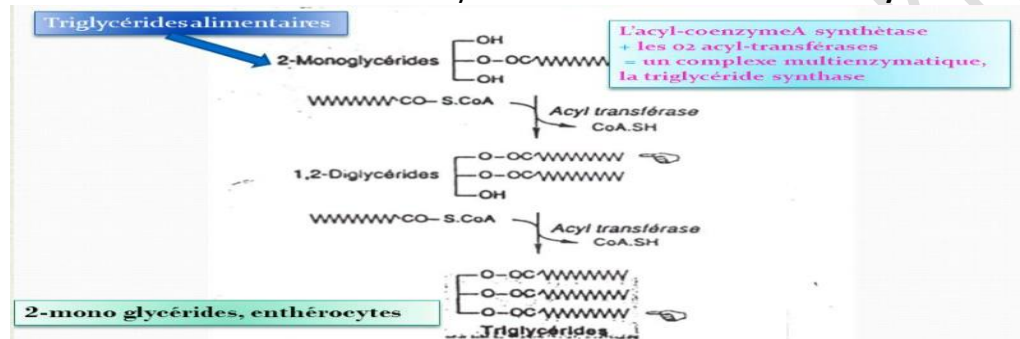


B/BIOSYNTHESIS REACTION:

❖ Exogenous TG synthesis from 2-MG:

In the intestinal lumen, the hydrolysis of dietary TG by **pancreatic lipase** yields fatty acids and 2-MG.

These serve as substrates for neosynthesis under the action of **TG synthase**.



TRANSACYLASE

Acyl-CoA synthetase and the **two acyl transferase** form a multi-enzyme complex known as **TG synthase**.

These TG, known as exogenous because their fatty acids are of dietary origin, are incorporated into **chylomicrons**, which transport them via the lymph and then the blood to **adipose tissue**, the site of storage, and to the **muscles and myocardium**, the main sites of use of fatty acids .

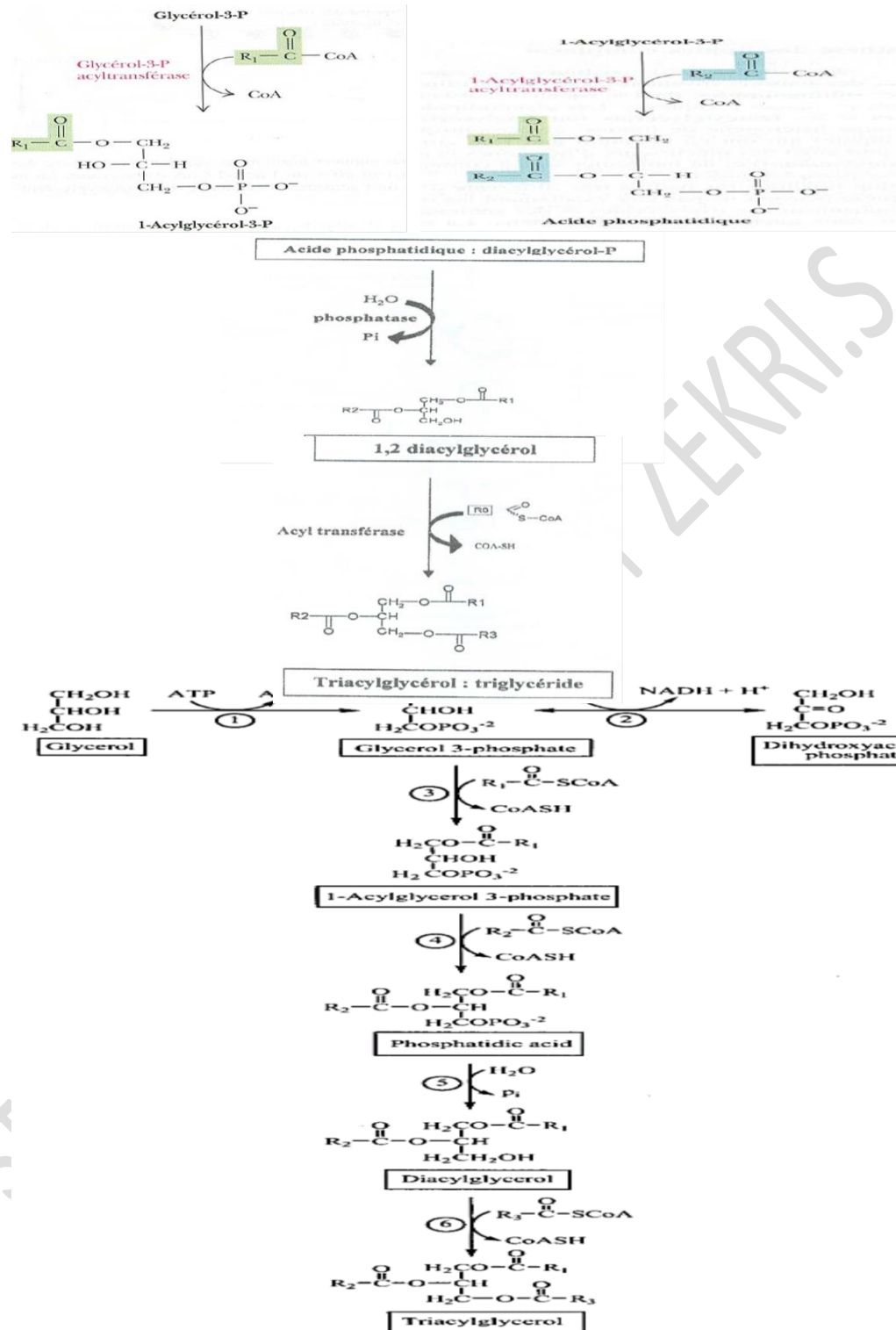
❖ Synthesis of endogenous TG from glycerol 3P

-In adipose tissue, the liver, muscles and myocardium,

-Thanks to **3 acyl transferases** and a **phosphatase** forming **TG synthase**

-Synthesis involves **three stages**:

- Formation of phosphatidic acid.
- Dephosphorylation of the latter to form diglyceride.
- Esterification of the terminal hydroxyl group of glycerol.



Biosynthesis of triglycerides

C/The regulation of lipogenesis is hormone-dependent:

- The uptake of carbohydrates into cells is facilitated by insulin, secreted by the pancreas.
- The enzyme triglyceride synthetase is activated by insulin.
- The enzyme triglyceride synthetase is inhibited by catecholamines such as adrenaline.

V. CATABOLISM OF TG:

1. Intestinal digestion:

-**Salivary and gastric lipases** (play an important role in newborns when pancreatic lipase activity is still low).

-**Pancreatic lipases**: hydrolyse TG into FAs + 2-MG.

Activated in the duodenum by trypsin and alkaline pH; essential cofactor: **colipase**.

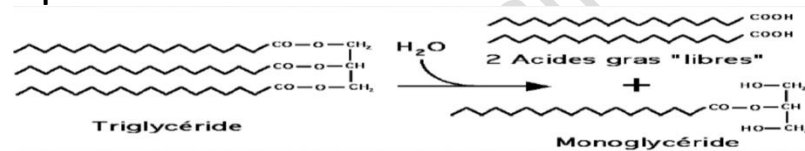
-Absorption in the form of micelles (with bile salts).

2. Breakdown of circulating triglycerides:

-**Lipoprotein lipase (LPL)**: Hydrolyses triglycerides from chylomicrons and VLDL into free fatty acids (used by tissues).

-**Apo CII** is an essential cofactor for its action

-**Apo CIII** is an inhibitor

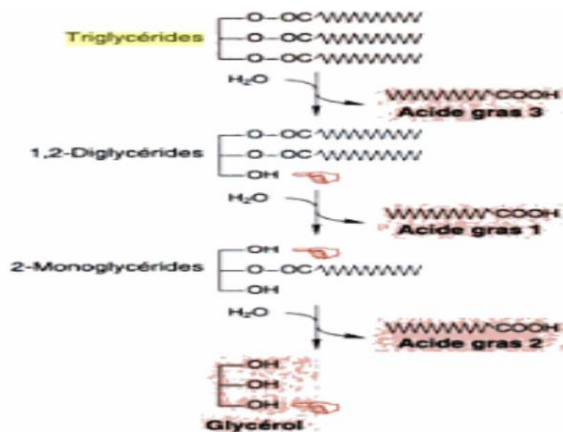


-Hepatic lipase: Breaks down triglycerides (TG) in IDLs and HDLs.

3. Mobilisation of reserves (adipose tissue):

-**Hormone-sensitive lipase (HSL):**

- The absence of glycerol kinase prevents the phosphorylation of glycerol, and thus the immediate resynthesis of triglycerides. Fatty acids produced by lipolysis therefore have no alternative but to be released from adipocytes.
 - Fatty acids in non-esterified form, bound to plasma albumin, are made available to the muscles and myocardium.
 - HSL is a rate-limiting enzyme in the regulation of TG metabolism
- Lipolysis contributes to thermogenesis, thanks to the energy released as heat during the hydrolysis of TG ester bonds.



VI. REGULATION OF TRIGLYCERIDE METABOLISM:

Lipogenesis and lipolysis coexist in a ^steady state ^ it is the interplay of supply and demand that determines the rate of either pathway.

❖ Hormonal regulation

LHS is the key enzyme in this regulation

- It exists in two forms:

-Active phosphorylated form

-Inactive dephosphorylated form

-Activated by phosphorylation (**glucagon, adrenaline**) → Releases fatty acids into the blood.

-Inhibited by dephosphorylation (**insulin**) → Promotes storage. It activates **protein phosphatase 2**, which acts by hydrolysing **cAMP**, there by **opposing lipolysis**.

Furthermore, **thyroid hormones, GH and glucocorticoids** are **lipolytic** by inducing LHS biosynthesis

❖ NUTRITIONAL AND ENERGY STATUS:

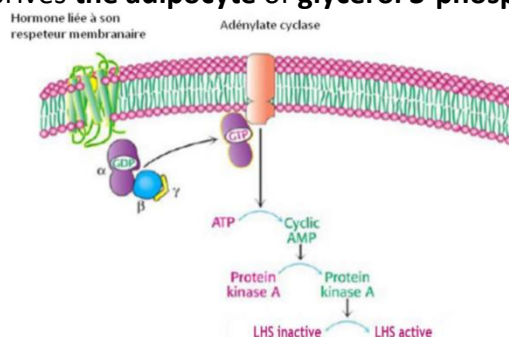
- **Post prandially:**

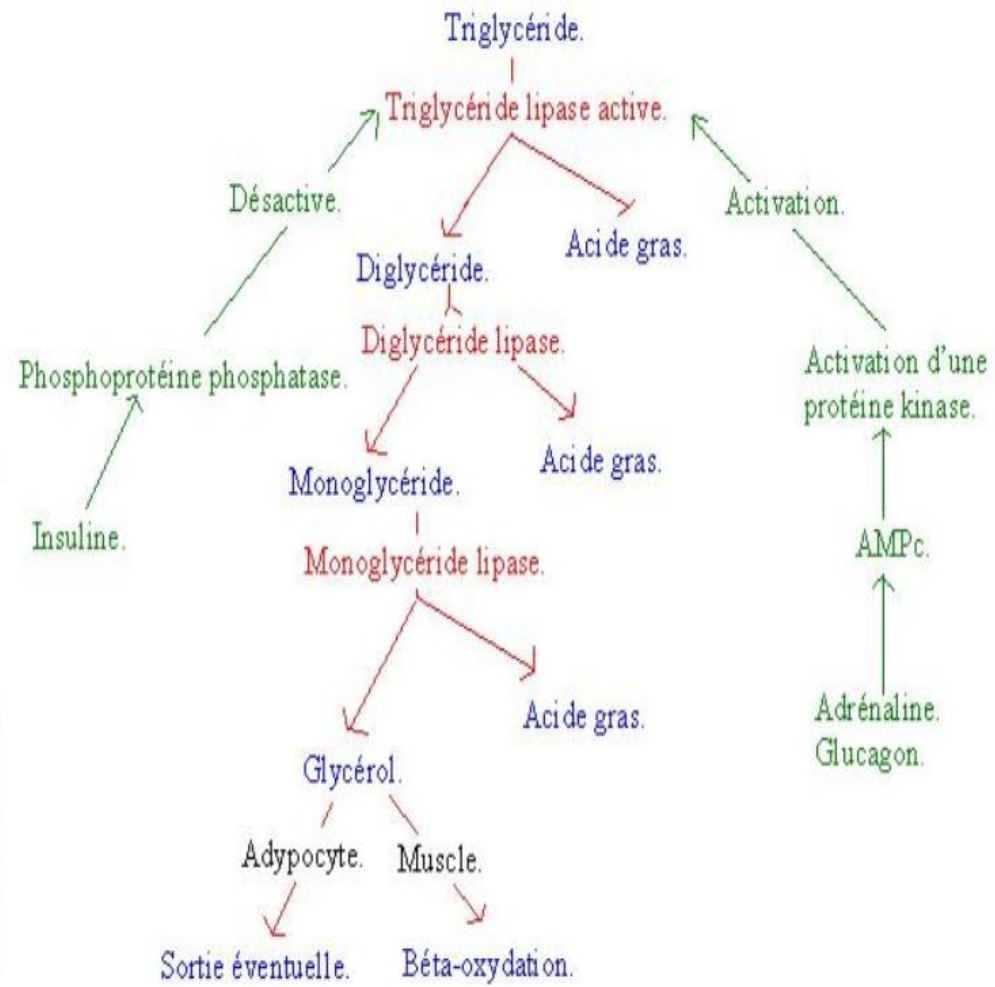
The increase in insulin and biosynthetic substrates promotes the biosynthetic pathway ==> adipose tissue replenishes these reserves

- **During fasting or muscular activity:**

Increased glucagon and/or adrenaline, along with the energy demand of energy-consuming tissues, promotes lipolysis.

Thus, in the postprandial period, **insulin** is **lipogenic**, whereas during fasting, the lack of insulin deprives the **adipocyte** of **glycerol 3-phosphate**.





LA LIPOGENOLYSE